

Bài tập lần 7,

Tài liệu Song - T-L - pdf Trang 18, 19.

Bài 3.

a) $u_{tt} = g u_{xx}$

$$u(x,0) = \chi_{[-1,1]}(x) = g(x) f(x)$$

$$u_t(x,0) = \chi_{[1,2]}(x) = g(x)$$

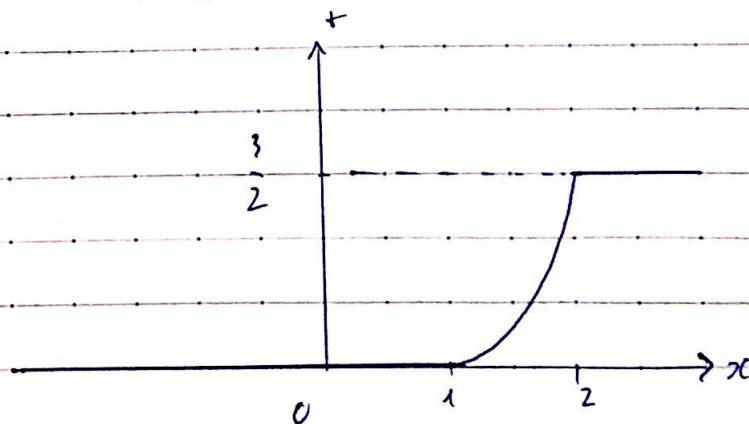
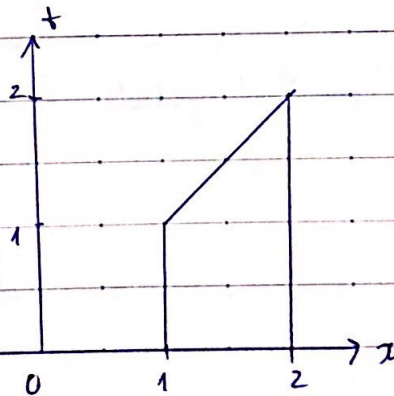
nghiệm của bài toán có dạng

$$u(x,t) = F(x-3t) + G(x+3t)$$

Song trên $F(x) = \frac{f(x)}{2} - \frac{1}{6} \int_0^x g(s) ds$

Song lưu $G(x) = \frac{f(x)}{2} + \frac{1}{6} \int_0^x g(s) ds$

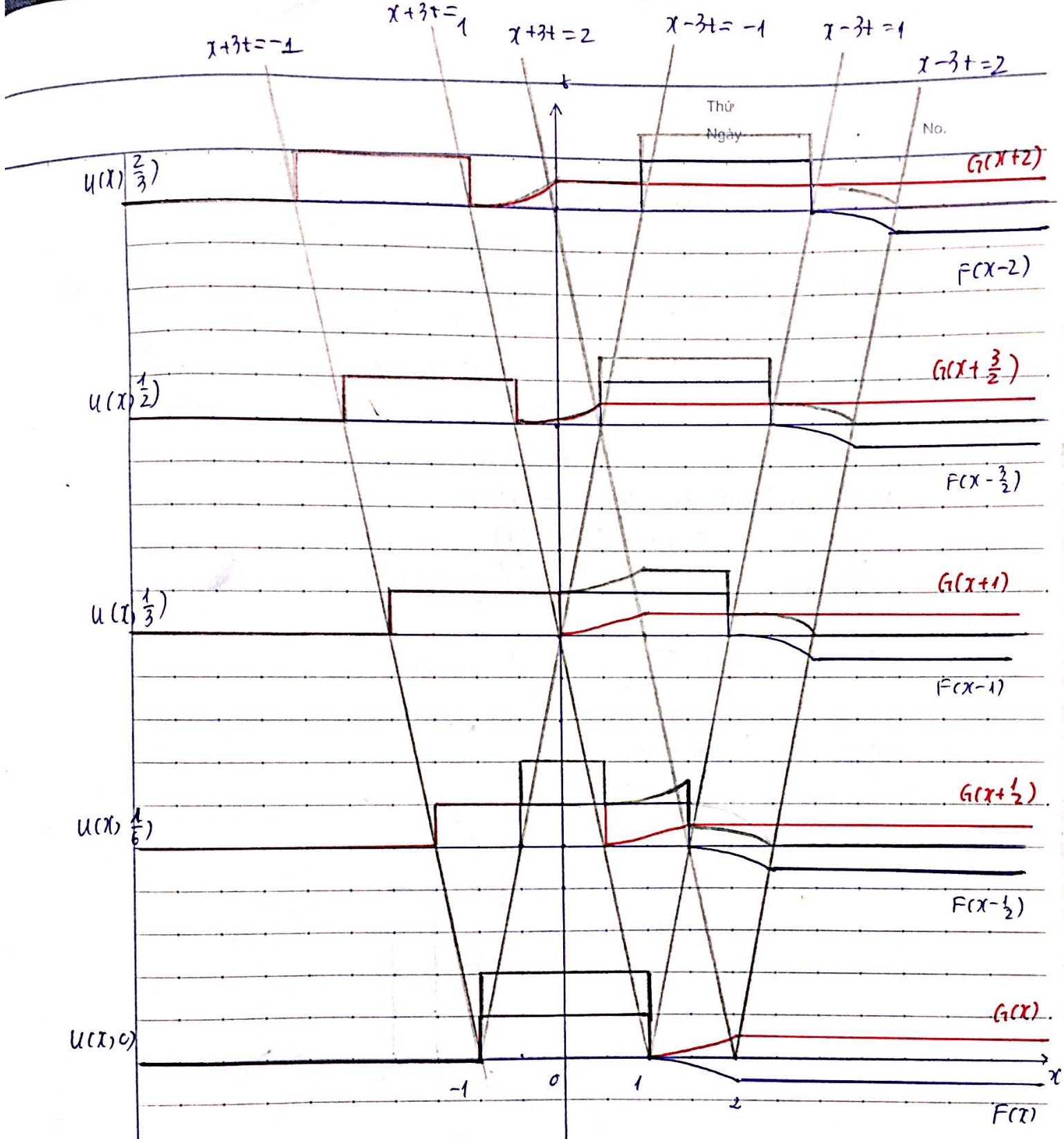
$$g(x) = \chi_{[1,2]}(x) = \begin{cases} x & \text{nếu } 1 \leq x \leq 2 \\ 0 & \text{nếu ngược lại} \end{cases}$$



$$\rightarrow \int_0^x g(s) ds = \begin{cases} 0 & \text{ne\u0169 } x < 1 \\ \frac{x^2}{2} - \frac{1}{2} & \text{ne\u0169 } 1 < x \leq 2 \\ \frac{3}{2} & \text{ne\u0169 } x > 2 \end{cases}$$

$$\rightarrow F(x) = \frac{f(x)}{2} - \frac{1}{6} \int_0^x g(s) ds = \begin{cases} 0 & \text{ne\u0169 } x < -1 \\ \frac{1}{2} & \text{ne\u0169 } -1 < x < 1 \\ -\frac{1}{6} \left(\frac{x^2}{2} - \frac{1}{2} \right) & \text{ne\u0169 } 1 < x < 2 \\ -\frac{1}{4} & \text{ne\u0169 } x > 2 \end{cases}$$

$$G(x) = \frac{f(x)}{2} + \frac{1}{6} \int_0^x g(s) ds = \begin{cases} 0 & \text{ne\u0169 } x < -1 \\ \frac{1}{2} & \text{ne\u0169 } -1 < x < 1 \\ \frac{1}{6} \left(\frac{x^2}{2} - \frac{1}{2} \right) & \text{ne\u0169 } 1 < x < 2 \\ +\frac{1}{4} & \text{ne\u0169 } x > 2 \end{cases}$$



Tập điểm kỳ dị của nghiệm gần 6 tia

$$\begin{cases}
 x+3t = \pm 1 \\
 x \pm 3t = 2 \\
 x-3t = \pm 1
 \end{cases}$$

Sách Pincovner - Rubinstein - Trang 93-97.

4.3

$$\begin{cases} u_{tt} - 4u_{xx} = 0 \end{cases}$$

$$\begin{cases} u(x,0) = \begin{cases} 1-x^2 & \text{nếu } -1 \leq x \leq 1 \\ 0 & \text{nếu ngược lại} \end{cases} = f(x) \end{cases}$$

$$\begin{cases} u_t(x,0) = \begin{cases} 4 & \text{nếu } 1 \leq x \leq 2 \\ 0 & \text{nếu ngược lại} \end{cases} = g(x) \end{cases}$$

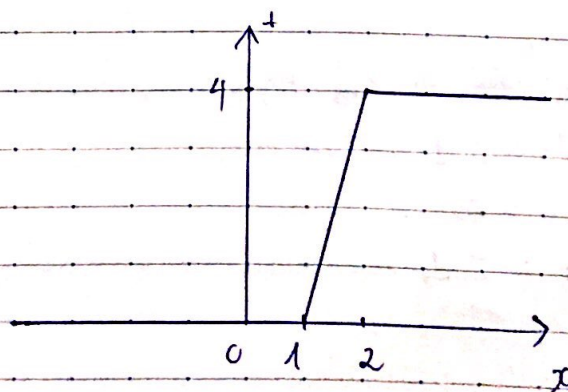
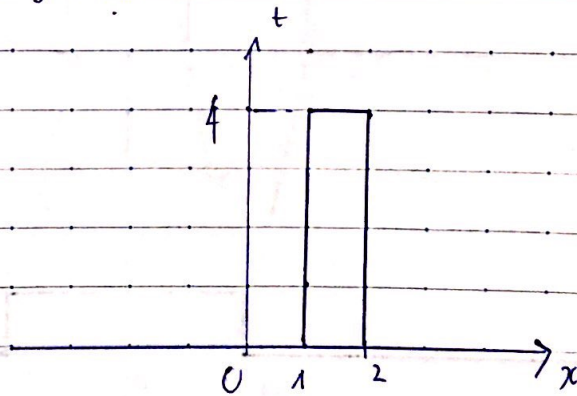
(a) Nghiệm của bài toán có dạng

$$u(x,t) = F(x-2t) + G(x+2t)$$

Song nên $F(x) = \frac{f(x)}{2} - \frac{1}{4} \int_0^x g(s) ds$

Song lùi $G(x) = \frac{f(x)}{2} + \frac{1}{4} \int_0^x g(s) ds$

$$g(x) = \begin{cases} 4 & \text{nếu } 1 \leq x \leq 2 \\ 0 & \text{nếu ngược lại} \end{cases}$$



$$\rightarrow \int_0^x g(s) ds = \begin{cases} 0 & \text{neũ' } x < 1 \\ 4x - 4 & \text{neũ' } 1 < x < 2 \\ 4 & \text{neũ' } x > 2 \end{cases}$$

$$\rightarrow F(x) = \frac{f(x)}{2} - \frac{1}{4} \int_0^x g(s) ds = \begin{cases} 0 & \text{neũ' } x < -1 \\ \frac{1-x^2}{2} & \text{neũ' } -1 < x < 1 \\ 1-x & \text{neũ' } 1 < x < 2 \\ -1 & \text{neũ' } x > 2 \end{cases}$$

$$G(x) = \frac{f(x)}{2} + \frac{1}{4} \int_0^x g(s) ds = \begin{cases} 0 & \text{neũ' } x < -1 \\ \frac{1-x^2}{2} & \text{neũ' } -1 < x < 1 \\ x-1 & \text{neũ' } 1 < x < 2 \\ 1 & \text{neũ' } x > 2 \end{cases}$$

$$x+2t=-1$$

$$x+2t=1$$

$$x+2t=2$$

Thứ

Ngày $x-2t=-1$

No. $x-2t=1$

$$x-2t=$$

$$G(x+3)$$

$$u(x, \frac{3}{2})$$

$$F(x-3)$$

$$G(x+2)$$

$$u(x, 1)$$

$$F(x-2)$$

$$G(x+1)$$

$$u(x, \frac{1}{2})$$

$$F(x-1)$$

$$G(x)$$

$$u(x, 0)$$

$$-1$$

$$0$$

$$1$$

$$2$$

$$F(x)$$

→ Tập đường đặc biệt của $u(x, t)$ gồm 6 tia

$$\begin{cases} x + 2t = \pm 1 \\ x \pm 2t = 2 \\ x - 2t = \pm 1 \end{cases}$$

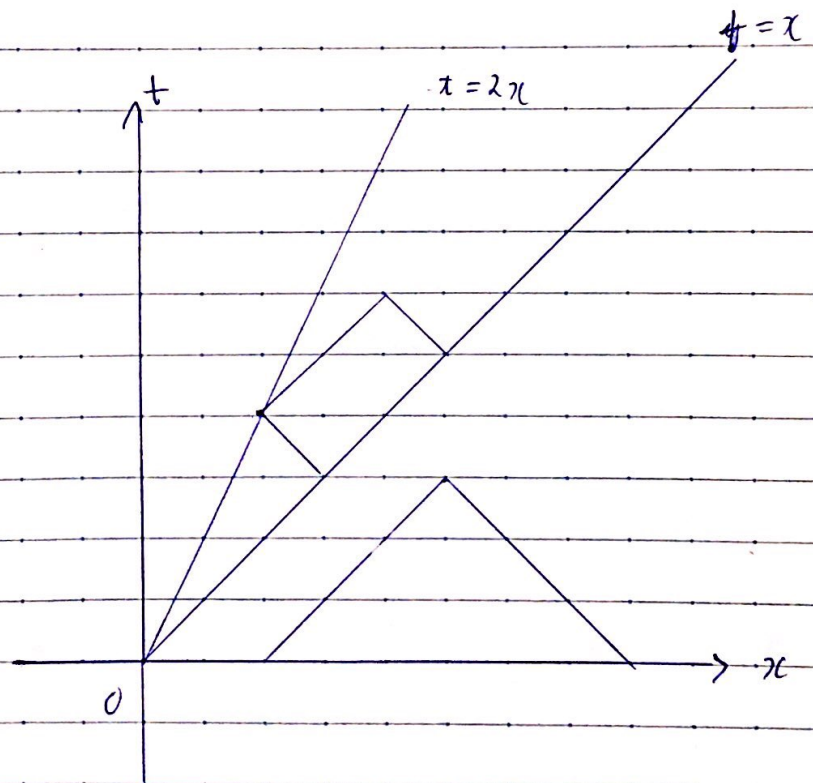
4.8.

$$u_t - u_{xx} = 0 \quad 0 < x < \infty, 0 < t < 2x$$

$$u(x, 0) = f(x) \quad 0 \leq x < \infty$$

$$u_t(x, 0) = g(x) \quad 0 \leq x < \infty$$

$$u(x, 2x) = h(x) \quad x \geq 0$$



→ Trong miền $0 < t < x$: Nghiệm của bài toán có dạng

$$u(x, t) = F(x-t) + F(x+t)$$

Sơ bộ nhận $F(x-t) = F(x) - \frac{f(x)}{2} - \frac{1}{2} \int_0^x g(s) ds$

Sơ bộ nhận $G(x) = \frac{f(x)}{2} + \frac{1}{2} \int_0^x g(s) ds$

$$u(x, t) = \frac{f(x-t) + f(x+t)}{2} + \frac{1}{2} \int_{x-t}^{x+t} g(s) ds$$

49

$$u_t - u_{xx} = 1 \quad x \in \mathbb{R}, t > 0 \quad (*)$$

$$u(x, 0) = x^2 \quad x \in \mathbb{R}$$

$$u_t(x, 0) = 1 \quad x \in \mathbb{R}$$

Phương trình đặc trưng

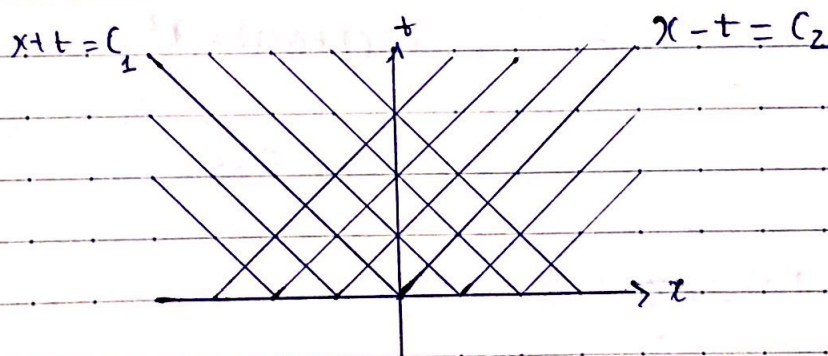
$$(dx)^2 - (dt)^2 = 0$$

$$\Rightarrow dx \pm dt = 0$$

$\rightarrow (*)$ là Hyperbolic

$$dx \pm dt = 0$$

$$\rightarrow \begin{cases} x+t = C_1 \\ x-t = C_2 \end{cases}$$



Đổi biến $\begin{cases} \xi = x+t \\ \eta = x-t \end{cases}$

		v_ξ	v_η	$v_{\xi\xi}$	$v_{\xi\eta}$	$v_{\eta\eta}$
0	u_x	1	1	0	0	0
0	u_t	1	-1	0	0	0
-1	u_{xx}	0	0	1	2	1
0	u_{xt}	0	0	1	0	-1
1	u_{tt}	0	0	1	-2	1
Tổng	1	0	0	0	-4	0

Dạng chung của (1) là

$$-4V_{\xi\eta} = 1$$

$$\Rightarrow V_{\xi\eta} = -\frac{1}{4}$$

$$\Rightarrow V_{\xi} = -\frac{1}{4}\eta + g(\xi)$$

$$\Rightarrow V = -\frac{1}{4}\eta\xi + F(\eta) + G(\xi)$$

$$\Rightarrow U(x,t) = -\frac{1}{4}(x-t)(x+t) + F(x-t) + G(x+t)$$

$$= -\frac{1}{4}(x^2 - t^2) + F(x-t) + G(x+t)$$

$$U(x,0) = -\frac{x^2}{4} + F(x) + G(x) = x^2$$

$$\Rightarrow F(x) + G(x) = \frac{5}{4}x^2$$

$$U_t(x,t) = \frac{t}{2} + G'(x+t) - F'(x-t)$$

$$\Rightarrow U_t(x,0) = G'(x) - F'(x) = 1$$

$$\Rightarrow \begin{cases} F(x) + G(x) = \frac{5}{4}x^2 & (1) \\ -F'(x) + G'(x) = 1 & (2) \end{cases}$$

$$\text{Đạo hàm (1)} \Rightarrow F'(x) + G'(x) = \frac{5}{2}x$$

Thay vào (2), ta được

$$-\left(\frac{5}{2}x - G'(x)\right) + G'(x) = 1$$

$$\Rightarrow G'(x) = \frac{5}{4}x + \frac{1}{2}$$

$$\Rightarrow G(x) = \frac{5}{8}x^2 + \frac{x}{2} + C_1$$

$$\rightarrow F(x) = \frac{5}{8}x^2 - \frac{x}{2} - C_1$$

$$\rightarrow u(x,t) = \frac{1}{4}(t^2 - x^2) + F(x-t) + G(x+t(t))$$

$$= \frac{1}{4}(t^2 - x^2) + \frac{5}{8}(x-t)^2 - \frac{x-t}{2} - C_1 + \frac{5}{8}(x+t)^2 + \frac{x+t}{2} + C_1$$

$$= \frac{1}{4}(t^2 - x^2) + \frac{5}{8}(x-t)^2 - \frac{x-t}{2} + \frac{5}{8}(x+t)^2 + \frac{x+t}{2}$$

$$= \frac{1}{4}(t^2 - x^2) + \frac{5}{8}(2x^2 + 2t^2) + t$$

$$= \frac{3}{2}t^2 + x^2 + t$$

Thử lại:

$$u(x,0) = x^2 \quad (T/M)$$

$$u_t(x,t) = 3t+1 \rightarrow u_t(x,0) = 1$$

$$\left. \begin{array}{l} u_{xx} = 2 \\ u_{tt} = 3 \end{array} \right\} \Rightarrow u_{tt} - u_{xx} = 1 \quad (T/M)$$